Advanced Image Processing Methods
Homework 4 Report

Peter Dobosi

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Goal

Estimate a relative depth (disparity) map from an uncalibrated stereo image pair using fundamental matrix estimation, uncalibrated rectification, and Semi-Global Block Matching.

Method

1. **Image Loading**: Both views (1920×1080) are loaded as grayscale with `cv2.imread`.
2. **Keypoint Detection & Matching**: SIFT is used for detection — its 128-D float descriptors yield more accurate sub-pixel matches than ORB, which is important since fundamental matrix quality depends directly on match accuracy. Descriptors are matched via FLANN (KD-Tree), filtered with Lowe's ratio test (0.75), and sorted by ascending distance. Matched (x, y) coordinates are returned as `float32` arrays.
3. **Fundamental Matrix**: Estimated with `cv2.findFundamentalMat` (`FM_RANSAC`, threshold 3.0, confidence 0.99). The returned mask is used to keep only inlier point pairs.
4. **Rectification**: `cv2.stereoRectifyUncalibrated` computes homographies H_L, H_R that align epipolar lines horizontally. Both images are warped with `cv2.warpPerspective`. Black border regions from uncalibrated rectification are expected and handled by SGBM.
5. **Disparity (SGBM)**: Computed with `cv2.StereoSGBM_create` (parameters in Table 1). The raw `int16` output (scaled by 16) is normalised to $[0, 255]$ `uint8`.
6. **Visualisation**: A 1×3 Matplotlib figure shows the rectified pair and the colormapped disparity map (brighter $\approx$ closer).

Parameter	**Value**	**Reasoning**
`minDisparity`	0	Standard left-as-reference.
`numDisparities`	128	Wide depth range; divisible by 16.
`blockSize`	5	Sharp depth edges; smoothness term compensates.
`P1`	200	$8 \cdot 1 \cdot 5^2$; small disparity change penalty.
`P2`	800	$32 \cdot 1 \cdot 5^2$; larger discontinuity penalty.
`disp12MaxDiff`	1	Left/right consistency check.
`uniquenessRatio`	10	Reject ambiguous matches ($\geq$10%).
`speckleWindowSize`	100	Speckle noise filter window.
`speckleRange`	32	Max variation inside a speckle.
`mode`	`SGBM_3WAY`	Quality/runtime trade-off.

Table 1: SGBM parameters. P_1, P_2 follow the recommended $8cb^2/32cb^2$ scaling (c=1 channel, b=5 block size).

Results

Figure 1 shows the rectified pair and disparity map. The central image region is well aligned and produces a coherent depth map; black border artefacts are inherent to uncalibrated rectification.

`output/disparity_results.png`

Figure 1: Rectified left, rectified right, and SGBM disparity map (brighter $\approx$ closer).